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Assessing social media use using a cognitive-behavioural marker of addiction

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Introduction

- Social media use is commonly assumed to have addiction-like qualities which may underpin its popularity and risk for excessive use.
- But unlike gambling or substance use disorders, self-report measures of social media 'addiction' lack thorough validation.

- We must move beyond symptom similarity and assess neurocognitive processes implicated in addictions.
- Incentive sensitisation theory predicts that addiction is driven by motivational 'wanting', a dopaminergic process, vs. hedonic 'liking'.
- This could underpin putative differences in addictiveness between social media and other forms of media.

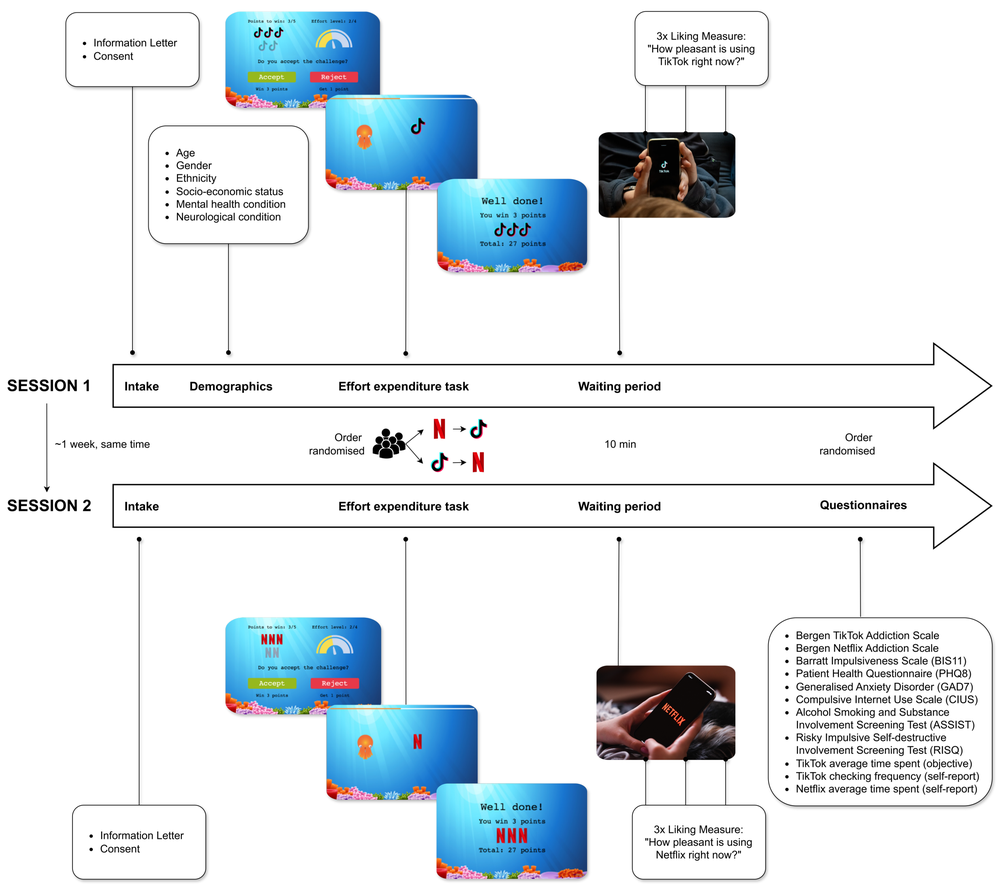
Hypothesis 1: TikTok is a more potent motivating stimulus than Netflix (α).

Hypothesis 2: The dissociation between *wanting* and *liking* is more pronounced for TikTok (α – liking).

Hypothesis 3: The dissociation between *wanting* and *liking* relates to self-reported addiction scores.

Methods

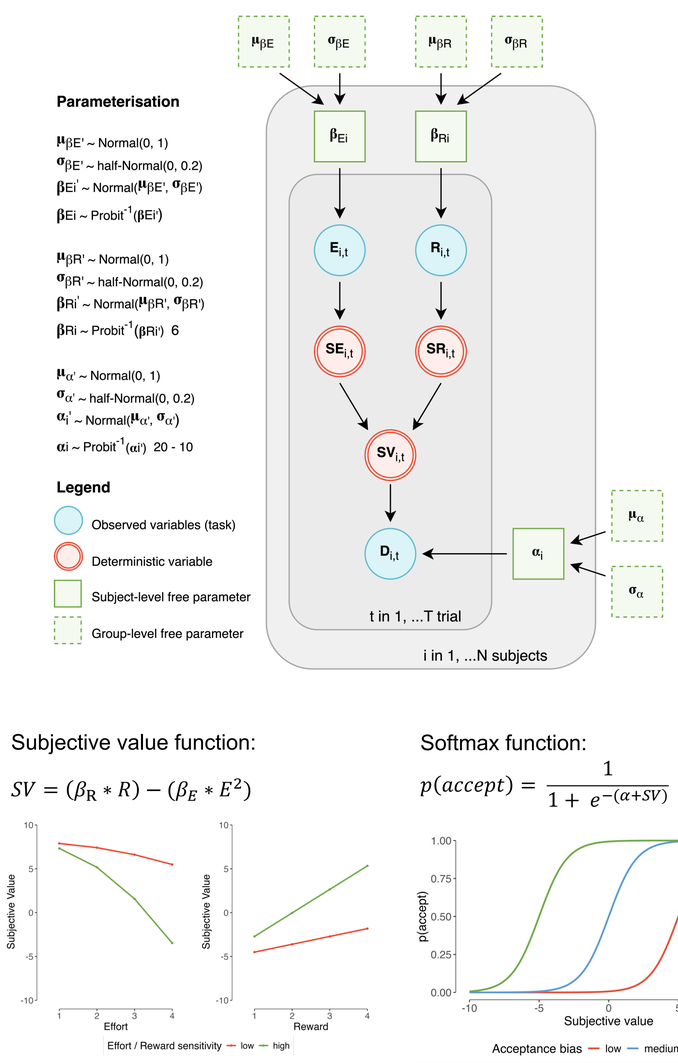
Experiment



Sample

N = 98 (F = 70), Age = 22.2 (18 – 27)

Effort-based Decision-making Model

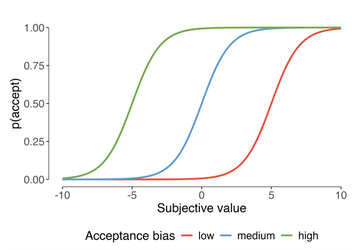


Subjective value function:

$$SV = (\beta_R * R) - (\beta_E * E^2)$$

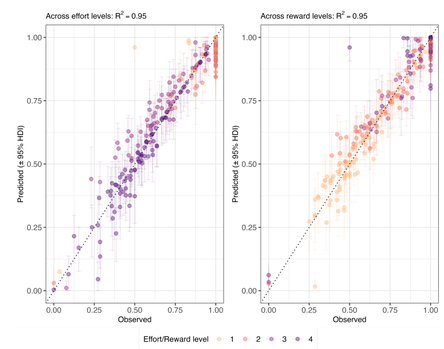
Softmax function:

$$p(accept) = \frac{1}{1 + e^{-(\alpha + SV)}}$$

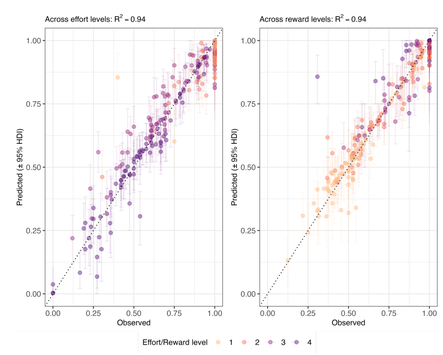


Model Validation

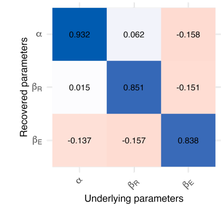
Posterior Predictive Check TikTok



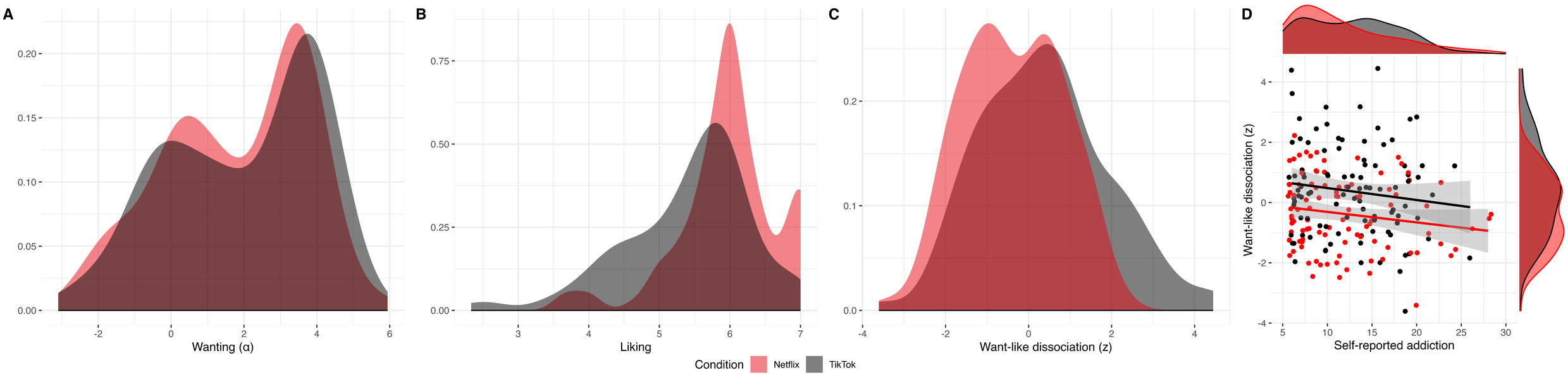
Posterior Predictive Check Netflix



Parameter Recovery



Results



Hypothesis 1: *BF_{01} = 0.178; 95% CI [-0.161, 0.448]*

Hypothesis 2: *BF_{01} = 121.107; 95% CI [0.350, 1.092]*

Hypothesis 3: *BF_{01} = 0.288; 95% CI [-0.259, 0.132]*

Discussion

- Contrasting Hypothesis 1, TikTok and Netflix are equally potent motivating stimuli (BF_{01} = 0.178).
- In line with Hypothesis 2, the dissociation of *wanting* and *liking* is more pronounced for TikTok, which may be an indication of an addiction-like process (BF_{01} = 121.107), the effect is driven by *liking*.
- Contrasting Hypothesis 3, The self-report measure of social media addiction and our cognitive-behavioural proxy do not correlate (BF_{01} = 0.288) which suggests that the two capture different things.
- To ensure we are not capturing general motivational tendencies we need to establish the sensitivity of the present paradigm to outcome differences, and longitudinal data would be valuable.
- More research is needed to validate predominant self-report measures of social media 'addiction' against the neurocognitive mechanisms established across decades of addiction research.

Pre-reg

References

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